

Introduction to Mathematics and Optimization

May, 2023

The references in the suggested solutions to this examination paper are with respect to the PDF edition *Introduction to Mathematics and Optimization* (Compressed static version (7.11.2022) of interactive book).

PROBLEM 1

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function

$$f(x, y) = x^4 + y^4 + xy + 1$$

and $U \subseteq \mathbb{R}^2$ the subset

$$U = \left\{ (x, y) \mid x > \frac{1}{3}, y > \frac{1}{3} \right\}.$$

- (a) Show that the complement $\mathbb{R}^2 \setminus U$ is a closed subset of $\mathbb{R}^2$ and that U is an open subset of $\mathbb{R}^2$.

ANSWER:

The complement is

$$\mathbb{R}^2 \setminus U = \left\{ (x, y) \mid x \leq \frac{1}{3} \vee y \leq \frac{1}{3} \right\},$$

which is a union of the two closed sets

$$\left\{ (x, y) \mid x \leq \frac{1}{3} \right\} = p_1^{-1}((-\infty, 1/3]) \quad \text{and} \quad \left\{ (x, y) \mid y \leq \frac{1}{3} \right\} = p_2^{-1}((-\infty, 1/3]),$$

where $p_1(x, y) = x$ and $p_2(x, y) = y$ are continuous mappings (Lemma 5.62) from $\mathbb{R}^2$ to $\mathbb{R}$. Conclusion via Proposition 5.52, Proposition 5.51, and Proposition 5.72. That U is open follows from Definition 5.53, since $\mathbb{R}^2 \setminus U$ is closed.

- (b) Show that $144x^2y^2 - 1 > 0$ if $(x, y) \in U$.

ANSWER:

Under the assumption, we have

$$144x^2y^2 - 1 > 144(1/3)^2(1/3)^2 - 1 = 144/81 - 1 > 0.$$

- (c) Compute the Hessian matrix for f and conclude that $f : U \rightarrow \mathbb{R}$ is a convex function. It may be used without proof that U is a convex subset of $\mathbb{R}^2$.

ANSWER:

The Hessian matrix is (according to Definition 8.2)

$$\begin{pmatrix} 12x^2 & 1 \\ 1 & 12y^2 \end{pmatrix}.$$

According to Exercise 3.41 and the previous part, this is positive definite for $(x, y) \in U$.

- (d) Show that $(0, 0)$ is a critical point for $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and that it is neither a local maximum nor a local minimum.

ANSWER:

According to Definition 7.17, the critical points are found by solving the equations

$$\begin{aligned}\frac{\partial f}{\partial x} &= 4x^3 + y = 0 \\ \frac{\partial f}{\partial y} &= 4y^3 + x = 0\end{aligned}$$

One sees that $x = 0 \iff y = 0$ in a solution. Thus $(0, 0)$ is a critical point. The Hessian matrix at $(0, 0)$ is

$$H = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This matrix is indefinite, since $uHu^t > 0$ for $u = (1, 1)$ and $vHv^t < 0$ for $v = (1, -1)$. Theorem 8.9 then states that $(0, 0)$ cannot be a local maximum or a local minimum.

- (e) Find all critical points $\neq (0, 0)$ for $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ and show that they are local minima.

ANSWER:

From the first equation, we have $y = -4x^3$ and hence $256x^8 = 1$ from the second equation, when $x \neq 0$. Thus $x = \pm \frac{1}{2}$. That is, for $x \neq 0$ and $y \neq 0$, the critical points are

$$\left(\frac{1}{2}, -\frac{1}{2}\right) \quad \text{and} \quad \left(-\frac{1}{2}, \frac{1}{2}\right).$$

These are substituted into the Hessian matrix. In both cases, it is

$$H = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix},$$

which is positive definite. Theorem 8.9 then states that they are local minima.

- (f) Show that

$$f(x, y) \geq \frac{7}{8}$$

for all $(x, y) \in \mathbb{R}^2$ (hint: You may use the substitution $x = r \cos(t)$ and $y = r \sin(t)$. It may be used without proof that $\cos(t)^4 + \sin(t)^4 \geq \frac{1}{2}$ for $t \in \mathbb{R}$).

ANSWER:

We use the substitution

$$\begin{aligned}x &= r \cos(t) \\ y &= r \sin(t)\end{aligned}$$

and obtain

$$f(x, y) = r^4(\cos(t)^4 + \sin(t)^4) + r^2 \cos(t) \sin(t) + 1 \geq \frac{1}{2}r^4 - r^2 + 1,$$

since $\cos(t)^4 + \sin(t)^4 \geq \frac{1}{2}$ for $t \in \mathbb{R}$ and $\cos(t) \sin(t) \geq -1$ for $t \in \mathbb{R}$. For $r \geq 2$, we have $f(x, y) \geq \frac{1}{2}r^4 - r^2 + 1 \geq \frac{1}{2}2^4 - 2^2 + 1 = 5$. For $r < 2$, we have $f(x, y) \geq \frac{7}{8}$ by Proposition 7.20, since f attains the value $\frac{7}{8}$ at the local minima from the previous problem. These local minima are $(-\frac{1}{2}, \frac{1}{2})$ and $(\frac{1}{2}, -\frac{1}{2})$, and both have $r < 2$.

PROBLEM 2

Consider (again) the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$

$$f(x, y) = x^4 + y^4 + xy + 1$$

and the open convex subset

$$U = \{(x, y) \mid x > \frac{1}{3}, y > \frac{1}{3}\}.$$

It is permitted to replace $\mathbb{R}^2$ with U when using results from the notes on convex optimization. From the previous problem, it is known that $f : U \rightarrow \mathbb{R}$ is a convex function. Let

$$F = \{(x, y) \mid x \geq \frac{1}{2}, y \geq \frac{1}{2}\}.$$

(a) Why does $F \subseteq U$ hold?

ANSWER:

If $x \geq 1/2$, then $x > 1/3$, since $1/2 > 1/3$. The same for y . Therefore $F \subseteq U$.

(b) Is $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ a convex function?

ANSWER:

No, it follows from PROBLEM 1 (d) that the Hessian matrix at the point $(0, 0)$ is

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

This matrix is not positive semidefinite, since

$$(xy) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2xy,$$

which is negative for $x = 1$ and $y = -1$. From Theorem 8.21, it follows that f cannot be convex.

(c) Show that F is a closed convex subset. Is F a bounded subset?

ANSWER:

The answer can be relatively simply modeled after the solution to PROBLEM 1 (a) and (c) from the examination paper January 2023. The subset F is not bounded according to §5.2.4, since $(x, x) \in F$ for $x \geq 1$ and $(x, x) \cdot (x, x) = 2x^2$ is unbounded for $x \geq 1$.

(d) Write down the KKT conditions for the convex optimization problem

$$\begin{aligned} \min & f(v) \\ & v \in F \end{aligned} \tag{1}$$

ANSWER:

The KKT conditions are

$$\begin{aligned}\lambda_1, \lambda_2 &\geq 0 \\ -x + \frac{1}{2} &\leq 0 \\ -y + \frac{1}{2} &\leq 0 \\ \lambda_1(-x + \frac{1}{2}) &= 0 \\ \lambda_2(-y + \frac{1}{2}) &= 0 \\ 4x^3 + y - \lambda_1 &= 0 \\ 4y^3 + x - \lambda_2 &= 0\end{aligned}$$

(e) Use the KKT conditions for (1) to show that $x = \frac{1}{2}$ and $y = \frac{1}{2}$ is a minimum.

ANSWER:

We substitute $x = y = 1/2$ into the KKT conditions. This gives

$$\begin{aligned}\lambda_1 &= 4(1/2)^3 + 1/2 > 0 \\ \lambda_2 &= 4(1/2)^3 + 1/2 > 0.\end{aligned}$$

Thus x, y together with the above λ_1, λ_2 give a solution to the KKT conditions. Therefore (x, y) is a minimum by Theorem 9.30 (ii).

(f) How many solutions do the KKT conditions for (1) have?

ANSWER:

If $(x, y, \lambda_1, \lambda_2)$ is a solution to the KKT conditions, (x, y) is an optimal solution to the optimization problem. An optimal solution is unique, since f is strictly convex on F : The Hessian matrix

$$\begin{pmatrix} 12x^2 & 1 \\ 1 & 12y^2 \end{pmatrix}.$$

is positive definite for $(x, y) \in U$ according to PROBLEM 1 (c) and Exercise 3.41. Conclusion by Theorem 8.21. Therefore, according to the previous part, there can only be one solution to the KKT conditions.

PROBLEM 3

Let A_t denote the 2×2 matrix

$$A_t = \begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix}$$

- (a) Show that A_t is positive definite if and only if $-1 < t$ and $t < 1$.

ANSWER:

This follows from Exercise 3.42, since $1 > 0$ and $1 - t^2 > 0$ for $-1 < t < 1$.

- (b) Show that A_t is positive semidefinite for $t \in \{-1, 1\}$. Here

$$A_{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \quad \text{and} \quad A_1 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}.$$

ANSWER:

For

$$B = \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix}$$

we have

$$B^T A_1 B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},$$

which is positive semidefinite according to Exercise 8.26. Hence A_1 is positive semidefinite via Theorem 8.27. For the matrix A_{-1} , we use

$$B = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

and

$$B^T A_{-1} B = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Same conclusion.

- (c) Show that A_1 is not invertible.

ANSWER:

With

$$v = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

we have $A_1 v = 0$. Hence A_1 cannot be invertible according to the remark under Definition 3.22.

- (d) Show that

$$A_t^{-1} = \begin{pmatrix} 1/(1-t^2) & -t/(1-t^2) \\ -t/(1-t^2) & 1/(1-t^2) \end{pmatrix}.$$

for $t \notin \{-1, 1\}$.

ANSWER:

Here one can explicitly verify that

$$A_t \begin{pmatrix} 1/(1-t^2) & -t/(1-t^2) \\ -t/(1-t^2) & 1/(1-t^2) \end{pmatrix} = \begin{pmatrix} 1/(1-t^2) & -t/(1-t^2) \\ -t/(1-t^2) & 1/(1-t^2) \end{pmatrix} A_t = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

according to Definition 3.22. For example,

$$\begin{pmatrix} 1 & t \\ t & 1 \end{pmatrix} \begin{pmatrix} 1/(1-t^2) & -t/(1-t^2) \\ -t/(1-t^2) & 1/(1-t^2) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

since

$$\begin{aligned}\frac{1}{1-t^2} + t\frac{-t}{1-t^2} &= \frac{1-t^2}{1-t^2} = 1 \\ \frac{-t}{1-t^2} + t\frac{1}{1-t^2} &= 0 \\ t\frac{1}{1-t^2} - \frac{t}{1-t^2} &= 0 \\ t\frac{-t}{1-t^2} + \frac{1}{1-t^2} &= \frac{1-t^2}{1-t^2} = 1.\end{aligned}$$

There is also the possibility of using the method in §3.5.1 or the content of Exercise 3.24.